

GANDAN HURYEE, Mongolia

WMO: 442210

Lat: 49.7006N

Lon: 96.3633E

Elev: 7156

StdP: 11.27

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -47.4	(c) -43.7	(d) -51.5	(e) 0.3	(f) -46.9	(g) -48.2	(h) 0.4	(i) -43.5	(j) 13.4	(k) -8.6	(l) 10.0	(m) -12.9	(n) 0.6	(o) 180	(p) 0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	83.4	57.3	79.8	56.1	76.6	54.9	60.8	73.1	58.9	72.7	57.4	71.4	6.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.6	89.4	65.5	54.3	82.2	63.7	52.3	76.2	62.8	30.6	73.1	29.1	72.8	27.9	71.4	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	13.6	11.7	-50.4	89.9	5.9	3.1	-54.6	92.2	-58.0	94.0	-61.3	95.8	-65.6	98.1
			-51.0	64.1	5.3	3.6	-54.8	66.7	-57.8	68.8	-60.8	70.8	-64.6	73.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.5	-26.1	-18.3	4.1	33.8	45.6	58.9	63.2	58.6	46.0	29.5	2.9	-18.1
	DBStd	33.00	10.25	11.71	13.03	11.34	8.10	6.75	5.41	6.20	7.81	10.01	13.01	10.37
	HDD50	10713	2358	1914	1424	489	183	11	0	9	167	635	1413	2110
	HDD65	15202	2823	2334	1889	937	602	200	97	212	570	1100	1863	2575
	CDD50	1057	0	0	0	2	46	277	410	275	47	0	0	0
	CDD65	72	0	0	0	0	0	16	42	14	0	0	0	0
	CDH74	1275	0	0	0	1	28	331	605	287	23	0	0	0
	CDH80	278	0	0	0	0	2	59	151	64	2	0	0	0
Wind	WSAvg	3.3	1.2	1.6	2.8	4.2	5.5	4.9	4.3	4.0	3.9	3.2	2.0	1.4
Precipitation	PrecAvg	7.6	0.2	0.1	0.2	0.3	0.4	1.2	2.1	1.7	0.7	0.3	0.2	0.3
	PrecMax	10.6	0.7	0.3	0.5	1.1	1.0	2.2	3.2	4.6	2.0	0.9	0.6	1.6
	PrecMin	5.2	0.0	0.0	0.0	0.0	0.0	0.2	0.6	0.2	0.2	0.0	0.0	0.0
	PrecStd	1.5	0.2	0.1	0.1	0.3	0.2	0.6	0.7	0.9	0.4	0.3	0.2	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	18.1	42.1	66.0	79.3	85.4	89.7	87.0	77.1	59.9	38.0	15.6
		MCWB	3.3	14.9	31.8	44.1	51.9	56.7	62.6	57.1	52.6	41.8	29.6	13.4
	2%	DB	-1.7	10.5	34.7	61.4	72.8	81.4	84.7	80.7	71.7	54.7	32.1	8.3
		MCWB	-2.9	8.8	29.4	42.3	49.6	55.9	59.0	56.1	50.6	39.9	27.2	7.1
	5%	DB	-6.0	4.8	29.8	56.6	68.5	77.8	80.6	76.7	66.9	49.9	27.6	3.5
		MCWB	-6.8	3.3	25.4	40.5	47.5	54.5	57.4	55.1	48.6	37.1	24.0	2.6
	10%	DB	-10.0	-0.1	25.0	51.9	64.1	74.4	77.0	73.0	62.1	45.6	22.3	-1.4
		MCWB	-10.6	-1.3	21.3	38.8	45.4	53.2	56.0	54.1	46.3	35.2	19.5	-2.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	14.9	33.3	46.4	54.7	60.8	65.3	61.2	55.0	43.2	31.0	13.7
		MCDB	4.5	18.2	40.0	60.3	72.4	74.5	82.2	72.6	70.8	57.1	36.9	15.7
	2%	WB	-2.7	8.8	29.4	44.1	50.9	58.8	62.1	58.9	52.3	40.8	28.3	6.9
		MCDB	-1.7	10.7	35.1	57.4	70.4	74.0	77.2	72.5	67.2	52.8	32.8	8.0
	5%	WB	-6.7	3.3	25.4	41.7	48.5	56.9	60.1	57.4	49.9	38.5	25.1	2.4
		MCDB	-6.0	4.8	29.7	54.7	66.1	71.5	72.7	71.4	64.2	49.2	29.1	3.2
	10%	WB	-10.6	-1.3	21.4	39.3	46.1	55.2	58.3	55.9	47.8	35.8	21.5	-2.3
		MCDB	-10.0	-0.2	24.9	51.6	62.2	69.9	71.8	69.4	60.3	45.8	24.5	-1.6
Mean Daily Temperature Range	5% DB	MDBR	17.7	21.8	25.8	22.2	25.4	24.3	22.2	23.7	26.0	24.2	19.2	16.7
		MCDBR	20.0	24.7	26.0	29.1	34.3	30.3	29.8	31.1	34.0	31.5	23.5	19.1
		MCWBR	19.2	22.8	20.4	13.3	16.7	11.8	11.2	12.3	16.0	18.1	15.1	18.1
	5% WB	MCDBR	20.2	24.4	24.7	25.8	30.6	23.8	22.8	24.4	29.4	29.7	18.6	19.2
		MCWBR	19.4	22.6	19.8	13.0	15.9	10.2	9.8	10.3	14.3	17.6	14.7	18.1
Clear-Sky Solar Irradiance	taub	0.203	0.224	0.256	0.291	0.299	0.301	0.305	0.290	0.260	0.245	0.226	0.207	
	taud	2.355	2.319	2.315	2.416	2.428	2.507	2.517	2.540	2.583	2.546	2.435	2.344	
	Ebn at Noon	285	302	306	305	305	304	301	302	304	292	272	266	
	Edh at Noon	20	27	32	33	35	32	31	29	25	21	18	17	
All-Sky Solar Radiation	RadAvg	494	823	1322	1758	2028	2080	1939	1689	1372	900	503	380	
	RadStd	23	38	46	75	102	82	97	81	84	54	36	21	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	-4.34	N/A	-1.58	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page